Honors Analysis Practice Midterm

October 13, 2025

Question 1 (40 points). *Write explicit examples for the following whenever possible. If not possible, simply state so. No proofs required.*

(a) A function that takes compact sets to compacts sets but is not continuous.

(b) A closed set that has no interior points.

(c) A set E for which $(\overline{E})^\circ \neq E^\circ$ (E° denotes the interior of E, i.e. the largest open subset of E).

(d) An open function $\mathbb{R} \to \mathbb{R}$ (with the usual topology) that maps a connected set to a disconnected set.

(e) A function $\mathbb{R} \to \mathbb{R}$ that is continuous at each irrational but discontinuous at each rational.

Question 2 (30 points). *Let p_n be a sequence in a metric space X with the following property: Every subsequence of (p_n) itself has a convergent subsequence which converges to p. Show that $\lim_{n\to\infty} p_n = p$.*

Question 3 (30 points). *A collection $\{V_\alpha\}$ of open subsets of a metric space X is said to be a* base *for X if the following is true: For every $x \in X$ and open set $G \subset X$ such that $x \in G$, we have $x \in V_\alpha \subset G$ for some α. In other words, every open set in X is the union of a subcollection of $\{V_\alpha\}$. Prove that every separable metric space has a countable base (a metric space is separable if there is a countable subset $Y \subset X$ such that $\overline{Y} = X$).*